

Policy Language and Information Effects in the Early Days of Federal Reserve Forward Guidance[†]

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I show that the nature of the Federal Open Market Committee's (FOMC's) forward guidance language shapes the private sector's responses to monetary policy statements. From February 2000 to June 2003, the FOMC only gave forward guidance about economic outlook risks, and a decrease in the expected federal funds rate path caused stock prices to fall, GDP growth forecasts to fall, and the unemployment rate to rise. From August 2003 to May 2006, the FOMC added forward guidance about policy inclinations, and a decrease in the expected federal funds rate path had the opposite effects. These results suggest that forward guidance that emphasizes economic outlook risks causes stronger information effects than forward guidance that emphasizes policy inclinations. (JEL D83, E52, E58, E66)

Central banks have increasingly used forward guidance—communication about the likely future course of interest rates and economic conditions—as a policy tool. Given its prominence, it is important to understand the effects of forward guidance.

Standard macroeconomic theory, where price stickiness is the primary economic friction, suggests that forward guidance influences current economic conditions by managing the expectations of private agents. Eggertsson and Woodford (2003) show that the entire path of expected future interest rates influences current economic activity and that lowering this path is expansionary. In this spirit, many researchers model forward guidance as future shifts in the central bank's policy rule that are announced in the present.¹ However, motivated by Romer and Romer's (2000) finding that the Federal Reserve has better inflation forecasts than private forecasters,

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¹Krugman (1998) is a seminal paper in this literature. More recent papers in this literature are Levin et al. (2010); Laséen and Svensson (2011); Werning (2011); Campbell et al. (2012); Milani and Treadwell (2012); Carlstrom, Fuerst, and Paustian (2015); Del Negro, Giannoni, and Patterson (2015); Bundick and Smith (2016); Campbell et al. (2017); and McKay, Nakamura, and Steinsson (2016).

an alternative theoretical literature treats information asymmetry as an important economic friction. In this literature, central bank policy reveals information to private agents or coordinates dispersed information. When these information effects are strong enough to dominate the standard effects of forward guidance, a lower interest rate path causes lower private-sector expectations for output growth and produces a contraction.² Because of this, forward guidance of lower future interest rates may be counterproductive at the zero lower bound (ZLB) on interest rates (Nakamura and Steinsson 2018).

This paper tests if the nature of the Federal Open Market Committee's (FOMC's) forward guidance language can affect whether the standard effects or the information effects of forward guidance are stronger. I do this by studying private sector responses to FOMC meeting statements from February 2000 to May 2006. I use this sample because it has an important change in the nature of forward guidance language. From February 2000 to June 2003, FOMC statements included language about the "risks of heightened inflation pressures or economic weakness in the foreseeable future." Notably, these statements had *no* direct language about future policy inclinations. However, on August 12, 2003, the FOMC stated, "policy accommodation can be maintained for a considerable period." This marked a change in FOMC statements to include direct language about future policy inclinations, and this direct language persisted until May 2006.

Using terminology from Kohn and Sack (2003), I say that the FOMC only used "economic-outlook" forward guidance from February 2000 to June 2003 and that it added "policy-inclination" forward guidance from August 2003 to May 2006. If adding policy-inclination forward guidance in August 2003 changed how the private sector interpreted FOMC statements, from simply giving guidance about economic conditions to committing to future monetary stimulus, then information effects may weaken and standard effects may be more relevant from August 2003 to May 2006 than from February 2000 to June 2003. Hence, I study FOMC statements from February 2000 to June 2003 and from August 2003 to May 2006 to see if the effects of forward guidance differed with the change in language.

I note several caveats to the hypothesis that this change in FOMC language would have changed the effects of forward guidance. First, policy-inclination forward guidance may not have communicated a material change in policy. If private agents expected a considerable period of economic slack and if a considerable period of low interest rates was consistent with the FOMC's past reaction function, then policy-inclination forward guidance may have been redundant. Second, FOMC forward guidance may not have been credible, and the private sector may have disregarded it. Related to this, if private agents disagreed with the FOMC's outlook but expected the FOMC to change future rates in accordance with that outlook, then standard theory may have applied with just economic-outlook forward guidance. Third, private agents may have taken policy-inclination forward guidance as an additional signal about the FOMC's outlook, causing strong information effects

²Ellingsen and Söderström (2001), Tang (2015), Gaballo (2016), Melosi (2017), Frankel and Kartik (2018), Nakamura and Steinsson (2018), Jia (2019), Andrade et al. (2019), and Jarociński and Karadi (2020) provide models where central bank policy reveals private information or coordinates dispersed information.

even if the FOMC intended to commit to future stimulus (Andrade et al. 2019). I address each of these caveats below.

The paper proceeds as follows. To address the first caveat in the previous paragraph, I begin by providing two forms of evidence that the August 2003 language change was a material change in policy. First, in Section I, I provide narrative evidence that the FOMC intended to communicate a break from prior policy in August 2003. Further, the FOMC's policy-inclination forward guidance was intended to lower expectations of future interest rates in the spirit of Eggertsson and Woodford (2003). Second, in Section II, I describe the federal funds rate surprises that will be used in the econometric model. They are changes in implied interest rates from federal funds futures contracts in narrow windows around FOMC meeting statements. I then provide empirical evidence that the magnitude of current federal funds rate surprises fell dramatically, beginning in August 2003. This fall is a break in the data that is consistent with the FOMC beginning to communicate its policy inclinations. While Section II follows in the spirit of Swanson (2006) and Swanson and Williams (2014), the result showing a structural break in federal funds rate surprises in August 2003 is novel.

Section III provides the main econometric model. It is an event-study regression that uses the federal funds rate surprises from Section II to measure two policy surprises. The first surprise is a change in the current federal funds rate. The second is an orthogonal change in the expected path of the federal funds rate, and I use this as the forward guidance surprise. I then regress changes in financial market variables during FOMC meeting days and changes in Blue Chip consensus forecasts during FOMC meeting months on the two policy surprises. I also adapt the model to regress changes in macroeconomic variables during the year following FOMC meetings on the policy surprises.

Section IV presents the main results of the paper. From February 2000 to June 2003, a surprise decrease in the orthogonal path of the federal funds rate caused immediate decreases in stock prices, Treasury yields, corporate bond yields, mortgage-backed securities (MBS) yields, and GDP growth forecasts, while also causing increases in term premia. It also caused lower inflation and a higher unemployment rate over the following year. These results suggest that FOMC statements that included only economic-outlook forward guidance had strong information effects. The financial market and Blue Chip forecast responses show that decreases in the expected path of the funds rate caused private agents to revise down their expectations for the economic outlook. The macroeconomic responses show that this downward revision of expectations was followed by worse economic conditions.

From August 2003 to May 2006, a surprise decrease in the orthogonal path of the federal funds rate caused immediate decreases in Treasury yields, corporate bond yields, and MBS yields, but immediate increases in stock prices with decreases in term premia, credit spreads, and the VIX. It also caused a subsequent increase in consumption and decrease in the unemployment rate. The FOMC added policy-inclination forward guidance in this sample, and the effects of forward guidance are consistent with standard macroeconomic theory. That is, a decrease in the expected path of the funds rate caused private agents to revise upward their outlook for the economy and generated an improvement in economic conditions.

Section V addresses possible confounding factors to my analysis. First, I provide evidence that although the FOMC began using policy-inclination forward guidance in August 2003 because interest rates were near the zero lower bound (ZLB), it was the language change and not the ZLB that generated my results. Second, I argue that although the February 2000 to June 2003 sample includes a recession and the August 2003 to May 2006 sample does not, there was not a clear change in economic uncertainty that would have changed how much private agents anchor their beliefs to the FOMC's outlook. In addition, I highlight August 9, 2011, as another FOMC meeting with an important change in forward guidance language. While policy-inclination language was not added or removed, it was meaningfully changed, and this meeting is an out-of-sample example consistent with the analysis in this paper.

Section VI concludes, and an online Appendix provides methodological details, additional and robustness analysis, and scatter plots of all data.

This paper shows that the effects of forward guidance can change when the FOMC changes the nature of its policy language. Three implications for monetary policy follow. First, the private sector does not ignore the FOMC's forward guidance, and both economic-outlook forward guidance and policy-inclination forward guidance appear credible. Second, the results for August 2003 to May 2006 suggest that private agents do not just treat policy-inclination forward guidance as a signal about the FOMC's economic outlook and that forward guidance can help manage ZLB episodes. Third, language about the economic outlook can reveal Federal Reserve information and may offset some effects of interest rate changes or policy-inclination language. While the standard theoretical effects of forward guidance dominated from August 2003 to May 2006, this might not always be the case.

This paper also addresses some puzzling results in the event-study literature. Similar to Gürkaynak, Sack, and Swanson (2005), I find that the orthogonalized path of the federal funds rate does not have a statistically significant effect on stock prices over the whole February 2000 to May 2006 sample. However, when I split the samples into February 2000 to June 2003 and August 2003 to May 2006 based on FOMC language, I find that the path of the funds rate has large effects on stock prices that flip signs. This suggests that the funds rate path is important for understanding the transmission of monetary policy shocks to equity markets, and that this transmission is time-varying. In addition, similar to Campbell et al. (2012) and Campbell et al. (2017), and Nakamura and Steinsson (2018), I find that an increase in the path of the funds rate increases private forecasts of inflation and GDP growth with decreases in forecasted unemployment rates in the February 2000 to May 2006 sample. Campbell et al. (2017) name these results the “event-study activity puzzle.” In this paper, these results are driven entirely by the February 2000 to June 2003 sample when the FOMC used only economic-outlook forward guidance. I find no such puzzle from August 2003 to May 2006, when the FOMC also used policy-inclination forward guidance.

This paper joins the literature on central bank communication. Blinder et al. (2008) provide a broad survey, and Rudebusch and Williams (2008) discuss FOMC language from 1999 to 2006. More specifically, I contribute to the event-study literature focused on FOMC communication, which includes Kohn and Sack (2003); Bernanke, Reinhart, and Sack (2004); Gürkaynak, Sack, and Swanson (2005); Campbell et al. (2012); Woodford (2012); Femia, Friedman, and Sack (2013);

Del Negro, Giannoni, and Patterson (2015); Gilchrist, López-Salido, and Zakrajšek (2015a); Hanson and Stein (2015); Sinha (2015); Campbell et al. (2017); Hattori, Schrimpf, and Sushko (2016); Bundick, Herriford, and Smith (2017); Swanson (2017); Nakamura and Steinsson (2018); and Lakdawala and Schaffer (2019); and the literature on the macroeconomic effects of FOMC forward guidance, which includes Gertler and Karadi (2015), Bundick and Smith (2016), Miranda-Agrippino (2016), D'Amico and King (2017), Lakdawala (2019), and Jarociński and Karadi (2020).³ In particular, I study Kohn and Sack's (2003) "economic-outlook" and "policy-inclination" aspects of forward guidance. Campbell et al. (2012) and Campbell et al. (2017) make a similar distinction between "Delphic" and "Odyssean" forward guidance: Delphic forward guidance forecasts economic performance and associated monetary policy actions, and Odyssean forward guidance publicly commits the FOMC to a future policy action.⁴ Distinct from all of the above papers, this paper's contribution is to use the FOMC's August 2003 language change to show that the strength of the standard effects and the information effects of forward guidance can depend on the nature of the FOMC's policy language.

This paper also relates to research that uses textual analysis. Lucca and Trebbi (2009) and Rosa (2011a) analyze FOMC language to indicate how the federal funds rate will change in the future. However, they do not separately classify that language into economic-outlook and policy-orientation aspects. Hansen and McMahon (2016) classify FOMC language into topics, but not into economic-outlook and policy-inclination topics. Hansen, McMahon, and Tong (2019) study the effects of central bank language on long-run interest rates.

I. Discussion of FOMC Statements

This section establishes that the nature of the FOMC's forward guidance changed materially in August 2003. First, I document that FOMC statements from February 2000 to May 2006 contained forward-looking language. Second, I argue that the forward-looking language from February 2000 to June 2003 was only about the economic outlook and avoided direct comments on policy inclinations. Third, I show that the forward-looking language changed in August 2003 to include the FOMC's policy inclinations. Further, FOMC minutes and transcripts indicate that this was a material change in policy intended to employ the standard theory of forward guidance. Fourth, I discuss ending the sample in May 2006.

Beginning with its February 2000 meeting, the FOMC provided forward guidance about economic risks. Importantly, those risks cover a time frame that extends *beyond* the next FOMC meeting. The FOMC announced this change in communication on January 19, 2000:⁵

³Rosa and Verga (2008) and Andrade and Ferroni (2016) study European monetary policy communication.

⁴I use Kohn and Sack's (2003) terminology because the forward guidance in my sample does not have the strong commitment characteristic that defines Odyssean forward guidance. In particular, the minutes of the August 12, 2003, FOMC meeting note that the FOMC "could not commit itself to a particular policy course." See <https://www.federalreserve.gov/fomc/minutes/20030812.htm>.

⁵See <https://www.federalreserve.gov/boarddocs/press/general/2000/20000119/default.htm> for the press release. I do not discuss forward guidance language prior to the February 2000 FOMC meeting because the Federal Reserve states that the FOMC "began using forward guidance in its post-meeting statements in the early 2000s." See <https://www.federalreserve.gov/faqs/what-is-forward-guidance-how-is-it-used-in-the-federal-reserve-monetary-policy.htm>.

[T]he FOMC changed its language describing its assessment of future developments. This new language will describe the FOMC's consensus about the balance of risks to the attainment of its long-run goals of price stability and sustainable economic growth and will be used in the announcement made after each meeting. More specifically, the announcement will indicate how the Committee assesses the risks of heightened inflation pressures or economic weakness in the foreseeable future. This time frame in the new language is intended to cover an interval extending beyond the next FOMC meeting.

Notably, the FOMC only provided this guidance about economic conditions and not about policy inclinations. The minutes of the December 21, 1999, FOMC meeting highlight this avoidance of the policy-inclination aspect of forward guidance:⁶

A few [FOMC] members wanted to retain the current focus on the possible future stance of policy ... The consensus opinion, however, was to replace the Committee's judgment about the likelihood of an increase or decrease in the intended federal funds rate with a description of the Committee's perception of the risks in the foreseeable future to the attainment of its long-run goals of price stability and sustainable economic growth.

Rudebusch and Williams (2008) note that, beginning in February 2000, "avoiding any references to future policy actions appeared important" and that "the taboo against any direct forward-looking signals about policy" was established. Following this, I view FOMC forward guidance in February 2000 as only containing an economic-outlook aspect. Hence, changes in private-sector federal funds rate expectations would have to be endogenous to the FOMC's outlook communication and based on its past reaction function. That is, the FOMC was not announcing shifts in its future policy rule as in, for example, the models of Campbell et al. (2012); Del Negro, Giannoni, and Patterson (2015); and Campbell et al. (2017).

The FOMC used economic-outlook forward guidance from February 2000 to May 2006. However, on August 12, 2003, the FOMC stated, "policy accommodation can be maintained for a considerable period." This was the first time in the sample that direct language about future policy inclinations appeared in a statement. Because of this, I use August 2003 as the first period that includes policy-inclination forward guidance.

An important issue is that this language change may not have reflected a shift in policy. If private agents had been expecting a considerable period of economic weakness, then a considerable period of policy accommodation may have been consistent with the FOMC's past reaction function. However, the August 2003 FOMC meeting minutes note, "the Committee would want to keep policy accommodative for a longer period than had been the practice in past periods of accelerating economic activity."⁷ Hence, the FOMC intended to communicate a break from past practice. Further, a longer period of policy accommodation despite accelerating

Consistent with this, Rudebusch and Williams (2008) note that the FOMC's policy tilt in its 1999 statements only applied to "the approximately six-week interval until the next meeting." Prior to 1999, they note that the FOMC's views on future policy were closely guarded and "only rarely discussed even internally."

⁶ See <https://www.federalreserve.gov/fomc/minutes/19991221.htm>.

⁷ See <https://www.federalreserve.gov/fomc/minutes/20030812.htm>.

economic activity follows the spirit of Eggertsson and Woodford (2003), making this language change useful for studying standard theories of forward guidance.⁸

The September 15, 2003, FOMC meeting, which immediately followed the August 2003 language change, provides additional evidence of the FOMC's policy change.⁹ Referring to the August 2003 statement, Donald L. Kohn said, "[T]he 'considerable period' sentence is a different matter. There we did make more of a semi-commitment about policy ... it was, in effect, a kind of nontraditional policy." Ben S. Bernanke followed-up with, "The words 'considerable period' were part of a nonstandard monetary policy. When we get close to the zero bound, we run out of traditional tools, and the only way that we can influence interest rates is by manipulating expectations." Hence, the FOMC was deviating from its previous policy and attempting to employ recommendations from standard macroeconomic theory.

The "considerable period" language began as a nonstandard policy to support the economy, but policy-inclination forward guidance was not immediately removed once the economy strengthened. Rather, it evolved over time. On January 28, 2004, the FOMC stated that it "can be patient in removing its policy accommodation." On May 4, 2004, it stated, "policy accommodation can be removed at a pace that is likely to be measured."¹⁰ On December 13, 2005, it stated, "some further measured policy firming is likely to be needed." On January 31, 2006, it stated, "some further policy firming may be needed." Finally, on May 5, 2006, it stated, "some further policy firming may yet be needed." The online Appendix has the forward-looking language for each FOMC statement from February 2000 to June 2006.

At the June 29, 2006, meeting, the FOMC stated, "[t]he extent and timing of any additional firming that may be needed ... will depend on the evolution of the outlook for both inflation and economic growth." It is not clear if I can interpret this language as being direct about policy inclinations. While the FOMC is direct about the direction of possible rate changes by noting additional firming, it is indirect by tying that firming to the economic outlook. Rudebusch and Williams (2008) similarly note that direct interest rate guidance was removed from this meeting's statement, and some participants at the June 2006 FOMC meeting indicated that scaling back direct interest rate guidance was their goal.¹¹ Because of this lack of clarity about how to categorize this language, I end the sample in May 2006.

While my sample period is motivated by the above changes in FOMC statement language, it has several additional appealing features. First, FOMC statements were

⁸FOMC meeting minutes are released with several weeks' delay. Hence, the private sector may not have immediately understood the intent of the FOMC's language change. However, Alan Greenspan's July 2003 testimony laid the groundwork for this language change by using the "considerable period" language. See <https://www.federalreserve.gov/boarddocs/hh/2003/july/testimony.htm>. Speeches by Ferguson (2003) and Bernanke (2003) in June and July 2003 also discussed how committing to low short-term interest rates could expand economic activity. Further, Bernanke (2003) specifically cited Eggertsson and Woodford (2003), indicating that standard macroeconomic theory was guiding the thinking of some FOMC participants.

⁹See <https://www.federalreserve.gov/monetarypolicy/files/FOMC20030915meeting.pdf>.

¹⁰The May 2004 language is important because it signaled that the FOMC would begin raising the federal funds rate and because the "measured" language persisted for a large part of the sample. However, this change in language did not reflect a reversion to pre-August 2003 policy. In the transcript of the May 2004 FOMC meeting, Vincent R. Reinhart said that the new language emphasized that the FOMC was "entering a tightening phase but one that likely will not be as aggressive as in prior episodes." See <https://www.federalreserve.gov/monetarypolicy/files/FOMC20040504meeting.pdf>.

¹¹See comments by Donald L. Kohn and Kevin Warsh on pages 98 and 127 of the June 2006 FOMC meeting transcript: <https://www.federalreserve.gov/monetarypolicy/files/FOMC20060629meeting.pdf>.

short and simple, relative to those during the ZLB period of December 2008 to November 2015 (Wynne 2013, Hernández-Murillo and Shell 2014), making forward guidance language easier to interpret. Second, summaries of economic projections (SEPs), which may include FOMC participants' statements about the appropriate monetary policy path or "dot plots" of the appropriate path, were not being released with FOMC statements and do not need to be interpreted. Third, the FOMC was not also announcing large-scale asset purchases (LSAPs). Hence, my sample has few confounding factors, allowing for clear interpretation of FOMC language. In addition, my sample has periods during which the target federal funds rate was low and constant. Thus, it provides evidence on how private agents react to forward guidance with pegged interest rates and is suggestive about the effects of forward guidance at the ZLB.¹²

II. Policy Surprises, Data, and a Structural Break

In this section, I measure changes in the expected federal funds rate path after an FOMC statement is released. These changes will be inputs in the econometric model in Section III. In Section IIA, I define these changes in expectations and describe the federal funds futures data that I use to measure them. In Section IIB, I show that the magnitude of the changes in the current federal funds rate fell beginning in August 2003. This break in the data is consistent with the FOMC directly communicating its policy inclinations, and it provides empirical evidence of a policy change in August 2003 to support the narrative evidence provided in the previous section.

A. Policy Surprises and Data

I begin with some notation. An FOMC meeting statement is released at moment t , r_t is the federal funds rate that is set in that statement, and $E_{t-\Delta_1}(r_t)$ is the expectation of r_t that is formed shortly before the statement's release. Then, the current federal funds rate surprise is

$$(1) \quad x_t^0 = r_t - E_{t-\Delta_1}(r_t).$$

Next, I measure how the current FOMC meeting statement changes expectations about the future path of interest rates. Let $E_{t-\Delta_1}(r_{t+1})$ be the expectation of the federal funds rate that will be set at the next FOMC meeting, where the expectation is formed shortly before the current FOMC meeting statement is released. Let $E_{t+\Delta_2}(r_{t+1})$ be the expectation of the federal funds rate that will be set at the next FOMC meeting, where the expectation is formed shortly after the current FOMC meeting statement is released. Then,

$$(2) \quad x_t^1 = E_{t+\Delta_2}(r_{t+1}) - E_{t-\Delta_1}(r_{t+1})$$

¹²From December 2001 to November 2002, the target federal funds rate was 1.75 percent. From November 2002 to June 2003, it was 1.25 percent. From June 2003 to June 2004, it was 1 percent.

is the change in expectations for the next FOMC meeting's interest rate caused by the current FOMC statement. Similarly, the change in expectations for the n th subsequent FOMC meeting's interest rate caused by the current FOMC statement is

$$(3) \quad x_t^n = E_{t+\Delta_2}(r_{t+n}) - E_{t-\Delta_1}(r_{t+n}).$$

To measure these changes in expectations, I use tick data from the 30-day Fed Funds Futures data from the CME Group's Time and Sales dataset (CME Group n.d.). From February 2000 to October 2003, I use "rth" or regular trading hours data from the trading floor. In December 2003, the "eth" or electronic trading hours data from the electronic trading platform become available, and I merge the price ticks for the rth and eth datasets. I measure x_t^0 , x_t^1 , x_t^2 , and x_t^3 following the methods used in Kuttner (2001); Gürkaynak (2005); and Gürkaynak, Sack, and Swanson (2005). The online Appendix has details. The three meetings following the current meeting typically occur within five or six months, so that x_t^1 , x_t^2 , and x_t^3 represent changes in expected rates with horizons out to six months. I use this time horizon because price updates in the data occur at a relatively high frequency out to about six months.¹³ Further, Gürkaynak, Sack, and Swanson (2007) show that federal funds futures forecast the future path of the funds rate better than other financial instruments out to six months. Finally, FOMC participants viewed their forward guidance horizon to be roughly six months.¹⁴

As in Gürkaynak, Sack, and Swanson (2005); Gilchrist, López-Salido, and Zakrajšek (2015a); and Nakamura and Steinsson (2018), I use a 30-minute window around the release of FOMC statements to compute x_t^0 , x_t^1 , x_t^2 , and x_t^3 . I set $t + \Delta_2$ to be 20 minutes following the statement release, and $t - \Delta_1$ is 10 minutes before the release. Given this narrow window, I assume that these changes in expectations can be attributed entirely to the statement. Gürkaynak, Sack, and Swanson (2005) also suggest that this 30-minute window provides sufficient time for participants in the federal funds futures market to trade based on their reading of the statement and that a 60-minute window yields very similar policy surprises.¹⁵

Based on Section I, my sample is scheduled FOMC meeting days from February 2000 to May 2006. I do not use days with inter-meeting rate changes or announcements.¹⁶ One reason for this is pragmatic: the test for structural change in the following subsection uses Greenbook data that are not available for inter-meeting changes. A second reason is conceptual. Inter-meeting policy changes are rare and occur when the FOMC perceives the change in economic

¹³ Early in the sample, a few observations of x_t^3 are measured with the previous day's closing price.

¹⁴ For example, see William Poole's comments on page 19 of the September 15, 2003, FOMC meeting transcript: <https://www.federalreserve.gov/monetarypolicy/files/FOMC20030915meeting.pdf>.

¹⁵ Following Gürkaynak, Sack, and Swanson (2005), I use the most recent trade price if there is no federal funds future trade exactly 10 minutes before an FOMC statement release. If there is no trade exactly 20 minutes following a statement release, I use the next available trade price. When there are multiple trades in the same minute, I average the prices within that minute to smooth out noise. Times for the releases of FOMC statements are taken from the Appendix of Lucca and Moench (2015). In the online Appendix, I also show that the main results in Section IV are robust to a longer window.

¹⁶ My sample has two inter-meeting rate changes that are sometimes included in other studies: January 3, 2001, and April 18, 2001. A third inter-meeting change occurred on September 17, 2001, which was the first day that New York markets opened after the September 11, 2001 terrorist attacks. This change happened at 8:20am, before the asset markets that I study opened. Hence, it is commonly excluded by other studies.

conditions to be unusually large. Hence, inter-meeting changes by their very nature may convey information about the economy independent of the FOMC statement language. Because inter-meeting announcements occur in the February 2000 to June 2003 sample but not in the August 2003 to May 2006 sample, they may lead to larger information effects from February 2000 to June 2003 independent of the August 2003 change in FOMC statement language. Hence, to best isolate the effects of the August 2003 language change, I only use scheduled FOMC meeting days in my baseline results. For robustness, I show results that include January 3, 2001, and April 18, 2001, in the online Appendix. They do not meaningfully change the forward guidance results presented in Section IV.

B. Empirical Evidence of a Structural Break

This subsection provides empirical evidence that the FOMC's language change in August 2003 was a material change in policy. In the spirit of Swanson (2006), I show that the magnitude of the current federal funds rate surprises, x_t^0 , fell beginning in August 2003.

Figure 1 shows x_t^0 for each scheduled FOMC meeting from February 1994 to May 2006.¹⁷ The gray region in Figure 1 covers the February 2000 to May 2006 sample studied in this paper, and the vertical dashed line separates the June 2003 and August 2003 FOMC meetings. Figure 1 shows that current federal funds rate surprises were much smaller in magnitude beginning with the August 2003 meeting. The average absolute value of x_t^0 from August 2003 to May 2006 was 0.6 basis points. In contrast, it was 4.6 basis points from February 2000 to June 2003, an average which was essentially the same as from February 1994 to December 1999. This large drop in current federal funds surprises is evidence that the FOMC's newly added policy-inclination forward guidance kept market participants informed of the FOMC's actual policy inclinations. Further, no such drop occurred when the FOMC began using its economic-outlook forward guidance in February 2000, indicating that the policy-inclination aspect of forward guidance was fundamentally different than the economic-outlook aspect.

Of course, many factors other than FOMC communication may affect current federal funds rate surprises. To better isolate the effects of communication on x_t^0 , I control for the following factors. First is the change in the target federal funds rate. Second, because recent changes in the federal funds rate may influence expectations for an upcoming change in the federal funds rate, I control for the change in the target funds rate from 90 days before an FOMC meeting to the day before an FOMC meeting, which is similar in spirit to Swanson's (2006) "momentum" variable. The third factor is the state of the business cycle. The fourth factor is how much the state of the business cycle has changed from the previous FOMC meeting. The online Appendix describes the data and their sources in detail.

¹⁷I use 1994 as the beginning of this sample because the FOMC began releasing statements with each policy change in 1994. My CME group data extend back only to 1995. For 1994, I use the monetary policy surprises from the Appendix of Gürkaynak, Sack, and Swanson (2005).

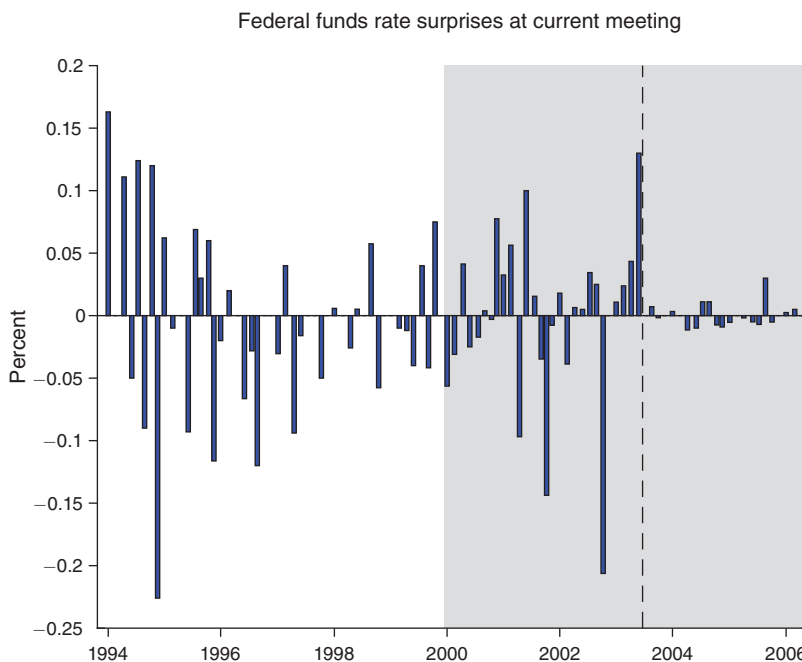


FIGURE 1. THE VALUE OF x_t^0 AT EACH SCHEDULED FOMC MEETING FROM FEBRUARY 1994 TO MAY 2006

Notes: The shaded gray region covers the meetings with forward guidance that are studied in this paper. The dashed vertical line separates the June 2003 and August 2003 meetings.

To control for these factors and formally test for a structural break in x_t^0 , I collect the relevant control variables into the vector w_t and estimate

$$(4) \quad x_t^0 = \delta_0 + w_t' \delta_1 + u_t.$$

I estimate equation (4) from February 1994 to May 2006. I use data from before the main sample in the paper for two reasons. The first is to examine whether a break occurred when the FOMC announced its economic-outlook forward guidance in 2000. The second is to provide for a sufficient number of observations given the vector of controls in equation (4).

Figure 2 shows the absolute value of the regression errors from equation (4). Even after controlling for the variables in w_t , current federal funds rate surprises were smaller beginning in August 2003. The average of $|\hat{u}_t|$ was 3.9 basis points from February 1994 to June 2003, but 1.8 basis points from August 2003 to May 2006. To test for a break in $|u_t|$, I estimate the average of $|\hat{u}_t|$ before and after each FOMC meeting from January 1997 to June 2004. I then construct a Wald statistic to test the null hypothesis that the two averages are the same. See the online Appendix for additional details.

Figure 3 shows the sequence of Wald statistics, which has a sharp spike at August 2003. The sup-Wald statistic of 17.2 in August 2003 exceeds Andrews's (1993) 1 percent critical value, rejecting the null hypothesis of no break in $|u_t|$. That is, the fall in the average magnitude of current federal funds rate surprises in August 2003

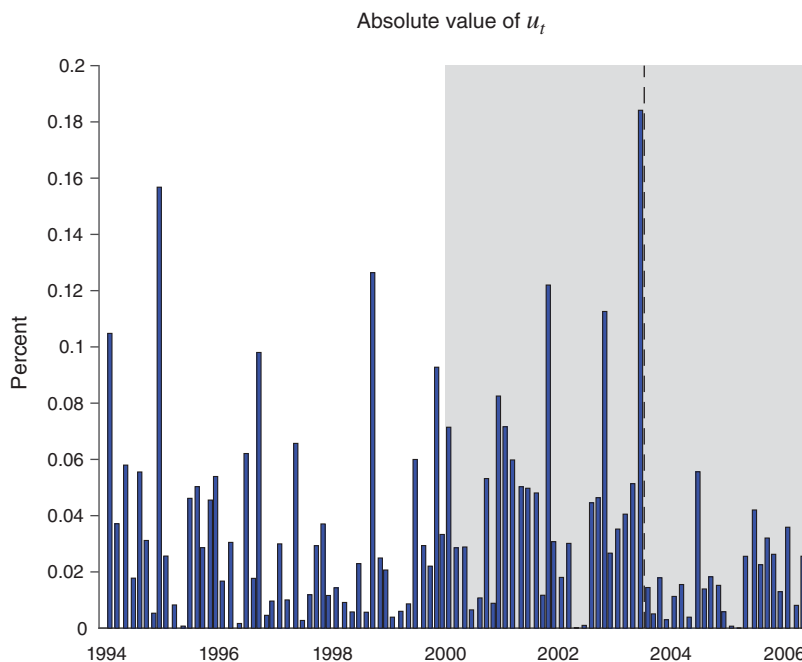


FIGURE 2. ESTIMATES OF $|u_t|$ FOR EACH SCHEDULED FOMC MEETING FROM FEBRUARY 1994 TO MAY 2006

Notes: The shaded gray region covers the meetings with forward guidance that are studied in this paper. The dashed vertical line separates the June 2003 and August 2003 meetings.

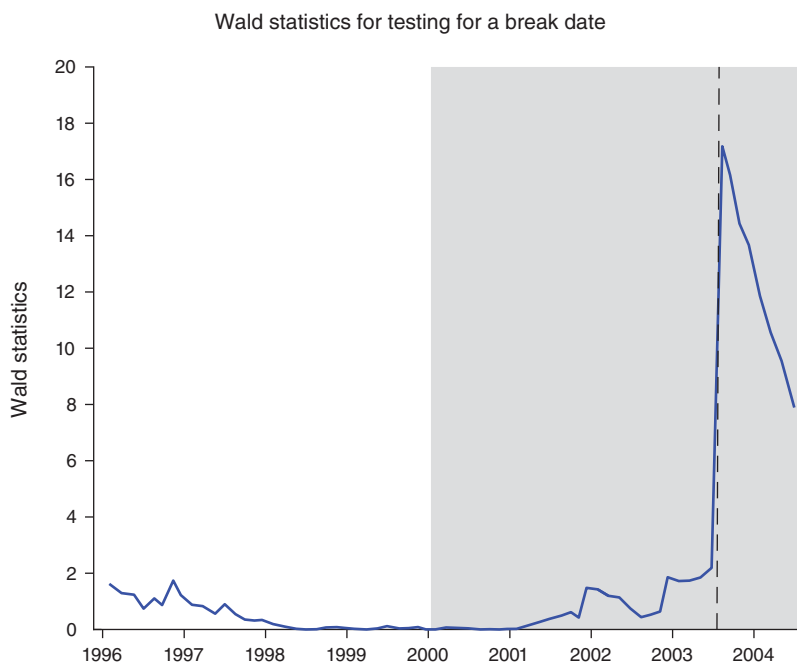


FIGURE 3. THE WALD STATISTIC FOR EACH SCHEDULED FOMC MEETING FROM JANUARY 1997 TO JUNE 2004

Notes: The shaded gray region covers the meetings with forward guidance that are studied in this paper. The dashed vertical line separates the June 2003 and August 2003 meetings.

is statistically significant, even if August 2003 is not treated as an *ex ante* break date. Hence, it is reasonable to treat the language change in August 2003 as a fundamental change in the FOMC's communication that kept market participants informed of the FOMC's actual policy inclinations.

The results here are similar to those in Swanson and Williams (2014). They found that 3-month Treasury yields and 1- to 2-quarters-ahead Eurodollar futures lost sensitivity to macroeconomic news, such as employment reports, during parts of 2002 to 2006. However, unlike the focus of this paper, they do not study the change in federal funds futures contracts around FOMC announcements. Hence, the results presented in this section are novel. Further, they found that 3-month Treasury yields began to lose sensitivity to macroeconomic news as early as 2002. In contrast, the spike in Figure 3 shows that August 2003 was the clear break in the magnitude of current federal funds rate surprises at FOMC meetings.

III. The Econometric Model

This section provides the main econometric model. I use an event-study model to estimate the immediate effects of monetary policy surprises on financial market variables and professional forecasts. I then adapt that model to estimate the subsequent effects on macroeconomic variables.

Following Gürkaynak, Sack, and Swanson (2005) and Gilchrist, López-Salido, and Zakrajšek (2015a), I create two orthogonalized policy shocks. The first policy shock is simply the surprise in the current federal funds rate, x_t^0 . The second policy shock is the forward guidance surprise. I estimate this shock in two steps. First, I use the simple average,

$$(5) \quad x_t^{path} = \frac{1}{3}(x_t^1 + x_t^2 + x_t^3),$$

to measure the change in the expected path of the funds rate. Second, I regress this change in the expected federal funds rate path on the change in the current federal funds rate,

$$(6) \quad x_t^{path} = \alpha_0 + \alpha_1 x_t^0 + m_t.$$

Here, m_t is the change in the expected path of the funds rate that cannot be predicted from and is orthogonal to the change in the current funds rate. I take m_t to represent the effects of forward guidance surprises on the path of the funds rate.

I estimate the effects of policy surprises on financial variables and forecasts with

$$(7) \quad y_t - y_{t-1} = \beta_0 + \beta_1 x_t^0 + \gamma m_t + e_t,$$

where y_t is the dependent variable of interest. Equations (6) and (7) compose the event-study model. Estimation and inference of this model uses generalized method of moments (Hansen 1982). Five moments, $E(m_t) = 0$, $E(x_t^0 m_t) = 0$, $E(e_t) = 0$, $E(x_t^0 e_t) = 0$, and $E(m_t e_t) = 0$, identify the five parameters of the model. Joint inference on equations (6) and (7) accounts for m_t being a generated regressor,

giving different standard errors than ordinary least squares, but not different point estimates. Details are in the online Appendix.

Next, I adapt this event-study model in order to estimate the macroeconomic effects of the monetary policy shocks. I modify equation (7) to be

$$(8) \quad y_{t+12} - y_t = \beta_0 + \beta_1 x_t^0 + \gamma m_t + e_{t+12},$$

where t now indexes a month, y_t is a macroeconomic variable of interest, and x_t^0 and m_t are the funds rate and forward guidance surprises from the FOMC meeting in month t . Equation (8) has two differences from equation (7). First, the dependent variable is a one-year change following an FOMC meeting. Second, the regression errors are now serially correlated, and heteroskedasticity and autocorrelation robust (HAR) inference is needed. I use a one-year horizon in order to give the macroeconomic variables time to respond to the policy surprises and to smooth through monthly noise, but this horizon is not so large that the regression errors are too persistent to account for given my sample sizes. An interpretation of equation (8) is that x_t^0 and m_t are orthogonalized structural shocks and that β_1 and γ are the cumulated impulse responses of the variable of interest to x_t^0 and m_t that are estimated by local projection (Jordà 2005). As with the event-study model, estimation and inference of equations (6) and (8) use GMM, and details are in the online Appendix.

Given the change in FOMC communication in August 2003, I estimate the models, equations (6) and (7), or equations (6) and (8), separately on the February 2000 to June 2003 and the August 2003 to May 2006 samples. The sample sizes are 28 and 23, respectively. Within each sample, estimates of m_t from equation (6) are the same for every dependent variable y_t . Figure 4 displays these estimates of m_t for the separate February 2000 to June 2003 and August 2003 to May 2006 samples. I relate these estimates of m_t to the FOMC's forward guidance language in the online Appendix, and I make three comments on these estimates here. First, m_t is not a measure of the stance of forward guidance; rather, it is a measure of market participants' surprise with forward guidance. Hence, interpreting estimates of m_t may involve assessing market expectations immediately prior to the release of FOMC statements. Second, the estimates of m_t generally correspond well to the FOMC's forward-looking language, providing reasonable measures of forward guidance surprises. As expected, m_t is driven by the economic outlook from February 2000 to June 2003, but it responds to both economic-outlook and policy-inclination language from August 2003 to May 2006. Third, forward guidance surprises were generally smaller in magnitude from August 2003 to May 2006 than in the earlier sample. This is natural because the FOMC's policy-inclination forward guidance included words such as "patient" and "measured" to describe potential policy changes. Hence, market participants did not expect large changes in the path of the funds rate, and the FOMC did not produce any.

IV. The Effects of Forward Guidance

This section presents the main results of the paper: empirical estimates of how financial markets, private forecasters, and macroeconomic variables reacted to forward guidance before and after the August 2003 language change. I estimate the

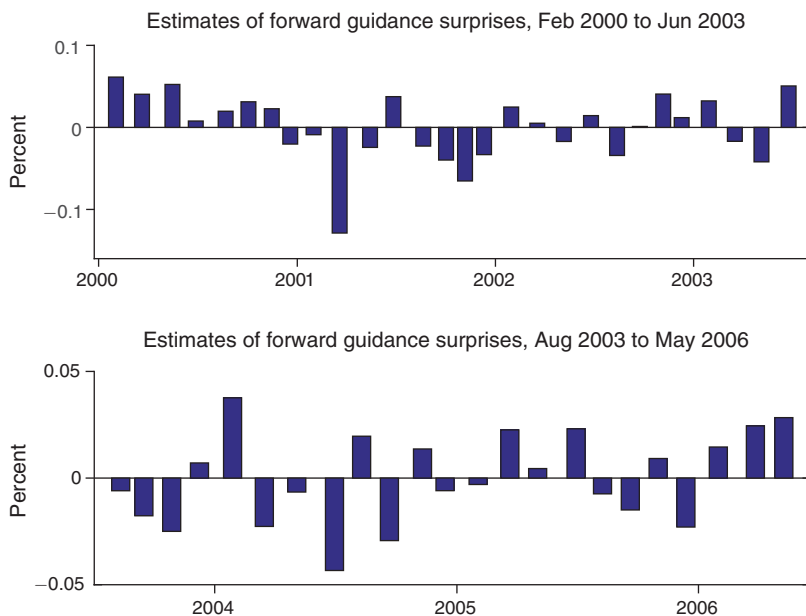


FIGURE 4. ESTIMATES OF FORWARD GUIDANCE SURPRISES

Note: Estimates are estimated separately on the February 2000 to June 2003 and the August 2003 to May 2006 samples.

coefficients, β_1 and γ , separately from February 2000 to June 2003 and from August 2003 to May 2006.¹⁸ I also estimate these coefficients over the whole sample to provide results comparable to those within the previous literature that does not focus on subsamples. Section IVA presents the effects of forward guidance on the stock market. Section IVB gives the effects of forward guidance on Treasury yields and term premia. Section IVC gives the effects of forward guidance on corporate bond rates and spreads as well as MBS rates and spreads. Section IVD shows the effects of forward guidance on Blue Chip forecasts. All results in Sections IVA to IVD use equations (6) and (7). Finally, Section IVE shows the effects of forward guidance macroeconomic variables. These results use equations (6) and (8).

A. The Stock Market

I first estimate the effects of current federal funds rate and forward guidance surprises on the stock market. While much of the related literature begins with the effects on Treasury yields, I begin with stock market results because they highlight

¹⁸Estimating the model separately on the two subsamples does not separately identify the effects of economic-outlook and policy-inclination forward guidance. Because the FOMC used only economic-outlook forward guidance from February 2000 to June 2003, the effects of this aspect are identified in this sample. However, the FOMC used both aspects of forward guidance from August 2003 to May 2006, so the model gives a combination of the economic-outlook and policy-inclination effects in this sample. I discuss one approach to separately identify the effects in the online Appendix and the data limitation that prevents it.

TABLE 1—RESPONSES OF STOCK PRICES AND VOLATILITY TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

Feb. 2000 to May 2006			Feb. 2000 to Jun. 2003			Aug. 2003 to May 2006		
Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2
<i>S&P 500 results</i>								
−7.88 (2.10)	4.08 (4.69)	0.18	−7.90 (2.07)	9.88 (4.29)	0.34	−8.08 (16.17)	−23.33 (6.51)	0.39
<i>VIX results</i>								
3.67 (1.70)	2.38 (4.55)	0.06	3.57 (1.72)	−2.22 (3.99)	0.09	9.53 (14.75)	20.78 (5.87)	0.30

Notes: The *Funds rate* columns display the estimates of β_1 and the *Forward guidance* columns display the estimates of γ from equation (7). Standard errors are shown in parentheses. See the text and the online Appendix for details about estimation and inference.

the differences in the two subsamples most clearly. I present the effects of forward guidance on Treasury yields next.

I use 100 times the natural logarithm of the S&P 500 to measure equity prices in percent. I also use the VIX as a measure of expected stock market volatility.¹⁹ Table 1 presents the estimates of β_1 and γ for stock prices and volatility. Standard errors are given in parentheses. Figures 5 and 6 show scatter plots of the data that underlie the S&P 500 results in Table 1. I provide scatter plots for all other dependent variables in the online Appendix. Because the forward guidance shocks are orthogonal from the current funds rate shocks, these two independent variables can be plotted separately without changing the regression coefficients.

Across all samples in Table 1, an increase in the current federal funds rate causes a decrease in stock prices. This finding is consistent with Bernanke and Kuttner (2005).²⁰ It also causes an increase in expected volatility. However, these effects are not statistically significant in the August 2003 to May 2006 sample. As discussed in Section IIB, the addition of policy-inclination language in August 2003 reduced the magnitude of the current federal funds rate surprises, and the bottom panel of Figure 5 shows that there is little variation in x_t^0 on which to regress stock price growth or any other dependent variable. The only observation of x_t^0 that takes a value greater than 0.01 percent is September 2005, which has a value of 0.03 percent and is the farthest right observation in the bottom panel of Figure 5. In the online Appendix, I use a leave-one-out analysis when estimating β_1 for stock prices to show that β_1 is sensitive to excluding September 2005. In the following subsections, I show estimates of β_1 for other dependent variables for the August 2003 to May 2006 sample for completeness. However, because forward guidance was essentially the only source of policy surprises from August 2003 to May 2006, caution is warranted when interpreting the values of β_1 in this sample.

¹⁹Daily S&P 500 prices are from the *Wall Street Journal*/Haver Analytics (Standard and Poor's n.d.). The VIX is from the Federal Reserve Bank of St. Louis's FRED database with database code VIXCLS (Chicago Board Options Exchange, n.d.).

²⁰The magnitude of my estimates are larger than Bernanke and Kuttner's (2005). They report an estimate of −4.68 in their Table II, column (b). However, my February 2000 to May 2006 estimate of −7.88 has a standard error of 2.10, putting Bernanke and Kuttner's estimate within my 90 percent confidence interval.

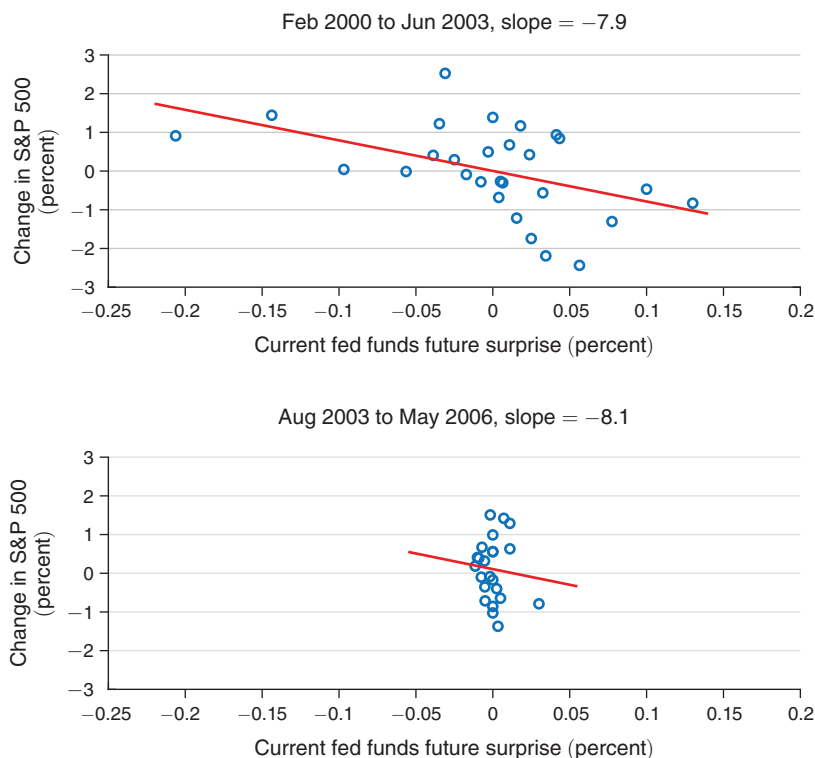


FIGURE 5. SCATTER PLOTS OF PERCENT CHANGE IN S&P 500 ON CURRENT FEDERAL FUNDS RATE SURPRISES

With regard to forward guidance for the whole sample, the effect on the stock market is smaller in magnitude than the effect of the current funds rate and is not statistically significant. Gürkaynak, Sack, and Swanson (2005) find a similar result, which they describe as “somewhat surprising.” They hypothesize that it is driven by investors revising their assessments of the economic outlook, consistent with information effects. My results support this hypothesis. From February 2000 to June 2003, an increase in the orthogonalized funds rate path causes a large increase in stock prices, which is consistent with investors revising up their own economic outlook as a result of economic-outlook forward guidance. However, from August 2003 to May 2006, an increase in the orthogonalized funds rate path caused a large decrease in stock prices, which is consistent with investors revising their expectations for the future stance of monetary policy as a result of policy-inclination forward guidance. These results indicate that stock price reactions to forward guidance can change, driven by changes in FOMC language, and create somewhat surprising results in larger samples.²¹

²¹I test the differences between the February 2000 to June 2003 and the August 2003 to May 2006 samples in the online Appendix. Following Gujarati (1970a, b), I perform a Chow (1960) test with dummy variables. The dummy variable is 0 from February 2000 to June 2003 and 1 from August 2003 to May 2006. For both dependent variables in Table 1, the differences between the effects of forward guidance in the August 2003 to May 2006 and the February 2000 to June 2003 samples are statistically significant at the 1 percent level.

This ability to resolve the stock market puzzle has important implications for the literature that has studied firm heterogeneity and the transmission of monetary policy by using the response of stock prices to monetary policy surprises. For example, heterogeneous pricing frictions (Gorodnichenko and Weber 2016), heterogeneous financial and information frictions (Ozdagli 2018), and production networks (Ozdagli and Weber 2017) are all shown to be relevant for monetary policy transmission. However, all of these papers focus on the response of stock prices to current federal funds rate surprises and do not study the responses to the expected path of the funds rate. Given the importance of forward guidance as a policy tool and given the large effects of forward guidance on stock prices in Table 1, research on firm heterogeneity and the transmission of forward guidance will be important for policy-makers. While this research is outside the scope of this paper, understanding which firm-level characteristics generate strong information effects versus standard theoretical effects may facilitate central bank communication with the public.

I note here that macroeconomic data releases, which commonly occur in the morning, do not materially impact my S&P 500 results. Given that my baseline specification uses daily percent changes in the S&P 500, such macroeconomic news may be a concern given my small sample sizes. However, in the online Appendix, I show that using percent changes in the S&P 500 with a roughly 2-hour window from 10 minutes before the FOMC statement until 4 PM gives very similar results to those in Table 1. In short, the S&P 500 does not appear to move much on the mornings before FOMC meeting statements in my sample. This low volatility prior to FOMC statements is consistent with Bomfim (2003) and Rosa (2011b). In my sample, essentially all of the change in the S&P 500 occurs in the afternoon after the release of FOMC statements, yielding very similar measures between daily and roughly 2-hour windows.

Figure 6 provides additional evidence on the robustness of my results. It shows scatter plots of stock price changes on forward guidance, and the change in the sign of the forward guidance effects from the early sample to the late sample is visually apparent. Hence, I can find statistically significant effects despite my small sample sizes. In the online Appendix, a leave-one-out analysis shows that the estimated effect of a forward guidance surprise is robust from August 2003 to May 2006. For the February 2000 to June 2003 sample, the March 2001 observation, which is the farthest left observation in the top panel of Figure 6, is influential when estimating the effect of a forward guidance surprise. Without it, the estimate of γ is 6.4 with a standard error of 6.0. That is, I still find a large and positive effect, but the estimate has less precision. Despite this, a Chow (1960) test shows that the difference between the effects of forward guidance in the August 2003 to May 2006 and the February 2000 to June 2003 samples is statistically significant at the 1 percent level, even without the March 2001 observation. In short, no single observation is responsible for generating the differing effects of forward guidance on stock prices shown in Table 1.

Jarociński and Karadi (2020) also use differing responses of stock prices to separate information effects from traditional monetary policy shocks. They measure surprises at FOMC meetings with the change in the three-month federal funds futures contract, which accounts for some near-term forward guidance. In a structural VAR, they find that an information shock that increases both interest rates and stock prices

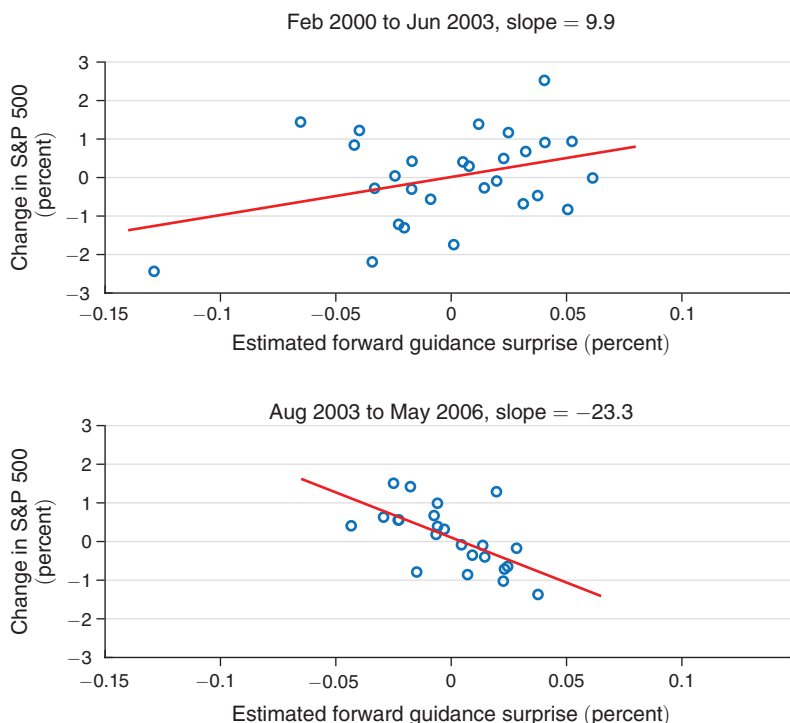


FIGURE 6. SCATTER PLOTS OF PERCENT CHANGE IN S&P 500 ON FORWARD GUIDANCE SURPRISES

also causes an increase in GDP. In contrast, a standard monetary policy shock that increases interest rates while decreasing stock prices causes a decrease in GDP. When combined with the Jarociński and Karadi (2020) results, Table 1 suggests that an increase in the federal funds rate path from February 2000 to June 2003 is expansionary but that an increase in the federal funds rate path from August 2003 to May 2006 is contractionary. Indeed, this is consistent with my results for Blue Chip forecast revisions and macroeconomic variables presented below.

Similar to the S&P 500, the VIX has different responses to forward guidance between the two samples. From February 2000 to June 2003, an increase in the orthogonalized funds rate path causes a decrease in the VIX, albeit small and not statistically significant. In contrast, from August 2003 to May 2006, an increase in the funds rate path causes a large increase in the VIX. Taken together, my results show that forward guidance that increases the funds rate path produced expansionary responses from the S&P 500 and the VIX from February 2000 to June 2003 but contractionary responses from August 2003 to May 2006.

B. Treasury Yields

Next, I estimate the effects of current federal funds rate and forward guidance surprises on the Treasury yield curve. I use Gürkaynak, Sack, and Wright's (2007)

continuously compounded zero-coupon yields as the measure of Treasury yields.²² In addition, I use Adrian, Crump, and Moench's (2013b) decomposition of the yield curve into term premia and the expectations of the future path of short-term nominal rates.²³

Table 2 presents the estimates of β_1 and γ for Treasury yields, term premia, and the expected path of short-term rates. All data are in percent so that a 1 percent change in the orthogonalized path of the funds rate causes a γ percent change in the dependent variable. Standard errors are given in parentheses. Over the whole sample and from February 2000 to June 2003, surprise changes in the current federal funds rate had an economically small and statistically insignificant effect on Treasury yields. These results seem to contradict the results in Gürkaynak, Sack, and Swanson (2005); however, my estimates all fall within one standard deviation of those in Gürkaynak, Sack, and Swanson, based on my standard errors. Further, as I show in the online Appendix, differences in data for the dependent variables and my choice to exclude inter-meeting changes can reconcile my results with Gürkaynak, Sack, and Swanson (2005).²⁴ Using changes in Treasury yields with a roughly 2-hour window (from 10 minutes before the FOMC statement until 4 PM) gives very similar funds rate results to those in Table 2, over the whole sample and from February 2000 to June 2003. I show this in the online Appendix. Hence, morning macroeconomic data releases are not influencing the funds rate results from February 2000 to June 2003.

In contrast to these results for the funds rate, forward guidance had a larger and statistically significant effect on 2-, 5-, and 7-year yields in the whole sample and the February 2000 to June 2003 sample. From August 2003 to May 2006, changes in forward guidance had very large effects on the yield curve, and the current funds rate had some importance for shorter yields. These results suggest that using policy-inclination forward guidance in addition to economic-outlook forward guidance generates a larger response from the yield curve than using economic-outlook forward guidance alone.²⁵

To gain further insight, I decompose the yield curve into term premia and the expectations of future short-term nominal rates. For the whole February 2000 to May 2006 sample, forward guidance has mostly small effects on term premia. However, as with the effects on stock prices, the whole sample hides important differences between the two subsamples. From February 2000 to June 2003, an increase in the orthogonalized funds rate path reduced term premia at longer horizons. In contrast, an increase in the orthogonalized funds rate path increased

²²Data are from <https://www.federalreserve.gov/pubs/feds/2006/200628/200628abs.html>, collected on October 25, 2018 (Board of Governors of the Federal Reserve System n.d.a).

²³Term premia data are from https://www.newyorkfed.org/research/data_indicators/term_premia.html, collected on October 25, 2018 (Federal Reserve Bank of New York n.d.). The expected future path of short-term yields is the Treasury yield at a given horizon less the corresponding term premium.

²⁴The insignificant changes in Treasury yields to federal funds rate changes in these samples seem to contradict the large observed stock price responses in Table 1. However, it is consistent with Bernanke and Kuttner (2005) who find that very little of the effect of funds rate changes on stock prices is directly attributable to changes in expected real interest rates.

²⁵The differences across the subsamples of the effects of forward guidance on the expected path of short-term rates are statistically significant at the 10 percent level or lower for each horizon. For every other dependent variable in Table 2, the differences between the effects of forward guidance are statistically significant at the 5 percent level or lower. See the online Appendix.

TABLE 2—RESPONSES OF TREASURY YIELDS AND TERM PREMIA TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

Feb. 2000 to May 2006			Feb. 2000 to Jun. 2003			Aug. 2003 to May 2006		
Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2	Funds Rate	Forward guidance	R^2
<i>Panel A. Treasury yields</i>								
2-year results								
0.26 (0.23)	1.23 (0.25)	0.43	0.21 (0.24)	0.92 (0.21)	0.37	3.15 (1.10)	2.29 (0.42)	0.68
5-year results								
0.13 (0.16)	0.90 (0.23)	0.30	0.10 (0.17)	0.61 (0.19)	0.21	2.04 (1.04)	2.05 (0.42)	0.61
7-year results								
0.11 (0.13)	0.70 (0.20)	0.22	0.08 (0.14)	0.42 (0.17)	0.12	1.43 (0.98)	1.83 (0.39)	0.56
10-year results								
0.10 (0.11)	0.48 (0.18)	0.13	0.09 (0.12)	0.22 (0.16)	0.05	0.88 (0.91)	1.57 (0.37)	0.49
<i>Panel B. Term premia</i>								
2-year results								
−0.08 (0.07)	0.11 (0.10)	0.05	−0.09 (0.07)	−0.02 (0.09)	0.05	0.55 (0.44)	0.74 (0.20)	0.40
5-year results								
−0.11 (0.10)	−0.15 (0.09)	0.06	−0.10 (0.10)	−0.27 (0.11)	0.16	−0.51 (0.50)	0.48 (0.23)	0.18
7-year results								
−0.09 (0.13)	−0.26 (0.10)	0.09	−0.08 (0.13)	−0.37 (0.14)	0.20	−0.89 (0.57)	0.38 (0.26)	0.14
10-year results								
−0.07 (0.16)	−0.34 (0.12)	0.10	−0.05 (0.16)	−0.47 (0.16)	0.21	−1.15 (0.65)	0.32 (0.30)	0.13
<i>Panel C. Expected path of short-term rates</i>								
2-year results								
0.34 (0.19)	1.12 (0.18)	0.61	0.30 (0.20)	0.95 (0.17)	0.62	2.60 (0.80)	1.56 (0.28)	0.67
5-year results								
0.23 (0.19)	1.05 (0.18)	0.54	0.20 (0.20)	0.88 (0.17)	0.53	2.56 (0.79)	1.56 (0.31)	0.65
7-year results								
0.20 (0.17)	0.96 (0.17)	0.53	0.16 (0.19)	0.80 (0.16)	0.51	2.32 (0.73)	1.45 (0.28)	0.65
10-year results								
0.17 (0.16)	0.82 (0.15)	0.51	0.14 (0.17)	0.69 (0.14)	0.50	2.03 (0.62)	1.25 (0.26)	0.64

Note: See notes to Table 1.

term premia at shorter horizons from August 2003 to May 2006. Adrian, Crump, and Moench (2013a) show that the 10-year term premium is countercyclical, and the term premia at other horizons are highly correlated. Thus, these term premia results are consistent with the stock market results in that a surprise increase in the funds rate path produced expansionary responses from February 2000 to June 2003, but contractionary responses from August 2003 to May 2006.

After removing the term premia to get the expectations of future short-term nominal rates, the effects of forward guidance become more similar across the

subsamples. However, the yield curve still responded between 60 to 70 basis points more to forward guidance from August 2003 to May 2006 than from February 2000 to June 2003. For the February 2000 to June 2003 sample, I note that forward guidance has more persistent effects on the expected path of short-term Treasury yields than one might expect from looking at the Treasury yield results alone. This is because the effects of forward guidance on term premia are negative in this sample, hiding the expectation effects of forward guidance.

C. Corporate Bonds and Mortgage-Backed Securities

I next estimate the effects of federal funds rate and forward guidance surprises on private-sector borrowing costs. Using data from Gilchrist, López-Salido, and Zakrajšek (2015a, b), I study the responses of corporate bond yields, MBS rates, and corporate bond and mortgage spreads. As they note, corporate bonds are often callable prior to maturity, and policy changes can affect the value of the call option. To abstract from this call option, I use their option-adjusted corporate bond yields based on the Bloomberg Fair Value model for A- and BBB-rated firms for maturities of 3 and 10 years. Yield spreads subtract the A yields from the BBB yields of comparable maturity. I also use Gilchrist, López-Salido, and Zakrajšek's (2015a) yield on the 30-year current coupon agency MBS and their option-adjusted spread (OAS) on the 30-year agency MBS. This spread is relative to the yield on comparable-duration Treasury securities and attempts to remove the option value of homeowners to prepay the mortgage in full. See Stroebel and Taylor (2012) for further discussion of OASs.

Table 3 presents the estimates of β_1 and γ for private borrowing costs. All data are in percent so that a 1 percent change in the orthogonalized path of the funds rate causes a γ percent change in the dependent variable. Standard errors are given in parentheses. For the February 2000 to June 2003 sample, the response of 3-year A-rated bonds to forward guidance is essentially the same as the response of 2-year Treasuries, and the response of 10-year A-rated bonds to forward guidance is similar to the response of 7-year Treasuries.²⁶ As with Treasury yields, A-rated bonds are more responsive to forward guidance in the August 2003 to May 2006 sample than in the February 2000 to June 2003 sample.²⁷

More important results for understanding the effects of forward guidance are the responses of BBB-rated bond yields and corporate yield spreads. In the February 2000 to June 2003 sample, BBB-rated bonds have essentially the same response as A-rated bonds to forward guidance, and the responses of corporate spreads are essentially zero. In contrast, BBB-rated bonds have a larger response to forward guidance than A-rated bonds in the August 2003 to May 2006 sample. The change in these spreads is economically large, with a 1 percent increase in the orthogonalized funds rate path causing a 0.38 percent to 0.64 percent increase in spreads, depending

²⁶I compare responses of the 3-year and 10-year A-rated bonds to the responses of the 2-year and 7-year Treasuries because coupon-bearing corporate bonds have shorter durations than the zero-coupon Treasury yields discussed above. I am grateful to Eric Swanson for pointing this out.

²⁷For the 3-year BBB – A spread, the difference across the subsamples of the effects of forward guidance is statistically significant at the 10 percent level. For every other dependent variable in Table 3, the differences are statistically significant at 5 percent levels or lower. See the online Appendix.

TABLE 3—RESPONSES OF PRIVATE-SECTOR BORROWING COSTS TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

Feb. 2000 to May 2006			Feb. 2000 to Jun. 2003			Aug. 2003 to May 2006		
Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2
<i>Panel A. Corporate bond and MBS yields</i>								
A 3-year results								
0.27 (0.20)	1.17 (0.24)	0.40	0.23 (0.22)	0.91 (0.24)	0.35	3.14 (1.01)	1.98 (0.39)	0.60
A 10-year results								
0.19 (0.17)	0.70 (0.19)	0.19	0.17 (0.18)	0.47 (0.20)	0.14	1.41 (0.90)	1.44 (0.38)	0.34
BBB 3-year results								
0.34 (0.17)	1.27 (0.24)	0.45	0.30 (0.18)	0.97 (0.17)	0.47	2.35 (1.42)	2.62 (0.55)	0.61
BBB 10-year results								
0.16 (0.16)	0.70 (0.22)	0.17	0.15 (0.17)	0.42 (0.21)	0.10	1.20 (1.13)	1.81 (0.48)	0.38
MBS 30-year results								
0.24 (0.12)	0.85 (0.21)	0.28	0.21 (0.13)	0.46 (0.15)	0.22	1.53 (1.26)	2.16 (0.42)	0.51
<i>Panel B. Corporate yield spreads and OAS</i>								
BBB – A 3-year results								
0.07 (0.09)	0.10 (0.13)	0.03	0.08 (0.09)	0.05 (0.13)	0.04	–0.79 (0.80)	0.64 (0.31)	0.31
BBB – A 10-year results								
–0.02 (0.07)	0.01 (0.12)	0.00	–0.02 (0.07)	–0.06 (0.13)	0.01	–0.21 (0.46)	0.38 (0.16)	0.24
OAS results								
–0.03 (0.05)	0.12 (0.07)	0.04	–0.04 (0.05)	0.00 (0.07)	0.01	0.69 (0.31)	0.54 (0.13)	0.44

Note: See notes to Table 1.

on duration. This suggests that economic-outlook forward guidance has little effect on interest rate spreads. However, policy-inclination forward guidance that raises the path of the funds rate also increases interest rate spreads—a contractionary effect.

The MBS yields and spreads show a similar pattern of responses. From February 2000 to June 2003, forward guidance had essentially the same effect on 30-year MBS yields as on the 10-year A-rated corporate bond yield. In addition, the response of the OAS on the 30-year agency MBS to forward guidance was essentially zero. However, from August 2003 to May 2006, the response of MBS yields to forward guidance was much larger than in the February 2000 to June 2003 sample, and the OAS on the 30-year agency MBS had a positive and large response to orthogonalized increases in the funds rate path.

D. Blue Chip Forecasts

I next estimate the effects of federal funds rate and forward guidance surprises on the consensus Blue Chip forecasts of GDP growth, CPI inflation, and the unemployment rate. These forecasts are made at a monthly frequency and are of quarterly averages. Following Nakamura and Steinsson (2018), the dependent variables

are changes in the average forecasted values over the next three quarters for every month with a scheduled FOMC meeting. At the beginning of the sample, the Blue Chip survey was conducted over three days, beginning on the first workday of each month. This was later shortened to two days. I remove observations where FOMC statements are released in these interview periods because I do not know if interview respondents observe the statement before responding. Nakamura and Steinsson (2018) make a similar adjustment. This leaves 44 total observations with 25 observations from February 2000 to June 2003 and 19 observations from August 2003 to May 2006.

Table 4 presents the estimates of β_1 and γ for private forecasts. All data are in percent so that a 1 percent change in the orthogonalized path of the funds rate causes a γ percent change in the forecast. Standard errors are given in parentheses. Before discussing the results, I note that statistical significance in Table 4 is sporadic, a result that is distinct from the previous tables. This is natural as I observe changes in financial variables on FOMC meeting days, while I only observe changes in Blue Chip forecasts on FOMC meeting months. Hence, more macroeconomic events can influence the Blue Chip forecasts relative to the financial variables, leading to more noise and less statistical power. However, with the understanding that the results in Table 4 are only suggestive, I still discuss the signs of the estimates in order to compare them to the previous literature and to my other results.

For the whole sample, an increase in the orthogonalized funds rate path causes increases in expected GDP growth and inflation and decreases in expected unemployment rates. The signs of these estimates are consistent with Campbell et al. (2012) and Campbell et al. (2017), and Nakamura and Steinsson (2018) and show that forward guidance that increases the funds rate path causes expansionary forecast revisions, not contractionary ones. Campbell et al. (2017) refer to these results as the “event-study activity puzzle.” Campbell et al. (2012) and Campbell et al. (2017) explain these results similar to how Gürkaynak, Sack, and Swanson (2005) explain their surprising stock price results. They argue that Delphic forward guidance reveals the Federal Reserve’s economic outlook and causes private forecasters to revise their own expectations. Based on Campbell et al.’s (2017) decomposition, the effects of this Delphic forward guidance appear to dominate the effects of Odyssean forward guidance, which publicly commits the FOMC to a future policy action. Nakamura and Steinsson (2018) have a similar story and provide a model wherein FOMC announcements affect expectations about both the stance of policy and the economic outlook.

My results are consistent with these explanations. From February 2000 to June 2003, an increase in the orthogonalized funds rate path caused increases in expected GDP growth and inflation and decreases in expected unemployment rates. This suggests that the FOMC’s economic-outlook forward guidance indeed causes private forecasters to revise their expectations in the same direction as the Federal Reserve’s outlook. These results are also consistent with the results in Tables 1, 2, and 3 that show that forward guidance that increases the funds rate path causes expansionary financial responses. Further, the sizes of the responses from February 2000 to June 2003 are similar to those in the whole February 2000 to May 2006 sample, suggesting that February 2000 to June 2003 is dominating the results.

TABLE 4—RESPONSES OF PRIVATE FORECASTS TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

Feb. 2000 to May 2006			Feb. 2000 to Jun. 2003			Aug. 2003 to May 2006		
Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2
<i>GDP Growth results</i>								
−0.04 (0.26)	1.11 (0.44)	0.12	−0.08 (0.26)	1.43 (0.46)	0.21	2.67 (2.72)	−0.82 (0.86)	0.12
<i>CPI Inflation results</i>								
−0.03 (0.29)	0.27 (0.28)	0.02	0.00 (0.26)	0.34 (0.33)	0.03	−2.23 (1.21)	−0.40 (0.57)	0.11
<i>Unemployment rate results</i>								
0.20 (0.24)	−0.56 (0.23)	0.07	0.20 (0.19)	−0.49 (0.26)	0.07	0.08 (0.74)	0.30 (0.49)	0.02

Note: See notes to Table 1.

Romer and Romer (2000), which motivates much of the literature on information effects, argues that movements in the federal funds rate reveal the Federal Reserve's information and influences private forecasts. In contrast, I find that the future path of the funds rate generates these information effects. However, I do not view my results as refuting the Romers' story. In a broad sense, both of our results show that the Federal Reserve can reveal information that influences private forecasts. More specifically, the difference in results is likely driven by changes in FOMC communication from the Romers' sample period, which covers 1974 to 1991, to my sample period. From 1974 to 1991, there were no FOMC meeting statements, and changes in monetary policy were revealed through open market operations, on which the *Wall Street Journal* regularly reported (Cook and Hahn 1989, Romer and Romer 2000). Hence, it is natural for the information content of the Fed's policy actions to migrate from open market operations and the corresponding changes in the federal funds rate to the FOMC's statements and their influence on longer-horizon futures contracts.

In more recent work, Nakamura and Steinsson (2018) show that monetary policy surprises have information effects on private forecasts from 1995 to 2014. However, they do not use separate funds rate and forward guidance surprises. Rather, they combine the two shocks by taking the first principal component of the unanticipated changes in several futures contracts. Because of this, I view my results as complementing theirs by highlighting the importance of forward guidance, relative to the funds rate in the 2000–2003 sample.

The results are different for the August 2003 to May 2006 sample, in which an increase in the orthogonalized funds rate path causes decreases in expected GDP growth and CPI inflation, and increases in the unemployment rate.²⁸ Campbell et al.'s (2017) event-study activity puzzle is not present in this sample. Rather, an increase in the orthogonalized funds rate path causes contractionary forecast

²⁸For GDP growth, the difference between the effects of forward guidance across the subsamples is statistically significant at the 5 percent level. For CPI inflation and the unemployment rate, the differences are not significant at the 10 percent level. See the online Appendix.

revisions, a finding that is consistent with the contractionary financial responses observed in Tables 1, 2, and 3. These results indicate that policy-inclination forward guidance can have standard theoretical effects.

E. Macroeconomic Variables

Lastly, I study the macroeconomic implications of forward guidance. To do this, I estimate the effects of federal funds rate and forward guidance surprises on real personal consumption expenditure (PCE) growth, CPI inflation, unemployment rate changes, and industrial production (IP) growth.²⁹ I use PCE and IP in this section because they are available monthly, while GDP is available only quarterly.

Table 5 presents the estimates of β_1 and γ for the macroeconomic variables. As noted above, these coefficients can be interpreted as cumulated impulse responses to a 1 percent increase in the funds rate or in the expected path of the funds rate. For HAR inference, I use the Bartlett kernel (Newey and West 1987) with a truncation parameter of 10. To assign levels of significance, I follow Lazarus et al. (2018) and use fixed- b critical values from Sun (2014).³⁰ As with the Blue Chip forecasts, statistical power is relatively low for these macroeconomic variables, and caution is warranted when interpreting the results. However, general patterns do emerge. First, for the whole sample, an activity puzzle is present as an increase in the path of the funds rate increases CPI inflation and decreases the unemployment rate while giving a positive coefficients for PCE and IP growth. These results are largely driven by the February 2000 to June 2003 sample, which has similar results. The one exception to this pattern from February 2000 to June 2003 is IP growth, which has a negative coefficient but is imprecisely estimated. The August 2003 to May 2006 sample shows a different pattern than the February 2000 to June 2003 sample: PCE growth falls and the unemployment rate rises in response to an increase in the path of the funds rate. Inflation is essentially unresponsive. The coefficient on IP growth is negative, which is consistent with PCE and the unemployment rate in this sample, and more precisely estimated than in the February 2000 to June 2003 sample.³¹

As discussed above, equation (8) can be interpreted as giving cumulated impulse responses to funds rate and forward guidance surprises. I choose a one-year change for the dependent variables in order to give the variables time to respond to the policy surprises and to smooth through monthly noise but without creating too much persistence in them. However, equation (8) can be generalized by using $y_{t+h} - y_t$ as the dependent variable where h can be interpreted as the impulse response horizon. I display these cumulated impulse responses to forward guidance surprises in

²⁹Real PCE is from the Bureau of Economic Analysis's National Income and Product Account Table 2.8.3, line 1 (US Bureau of Economic Analysis n.d.). CPI, the unemployment rate, and IP are from Federal Reserve Bank of St. Louis's FRED database with database codes CPIAUCSL (US Bureau of Labor Statistics n.d.a), UNRATE (US Bureau of Labor Statistics n.d.b), and INDPRO (Board of Governors of the Federal Reserve System n.d.b).

³⁰Lazarus et al. (2018) suggest that the truncation parameter be chosen with $1.3T^{1/2}$, where T is the sample size. For my samples, this implies truncation parameters between 6 and 9. Because my dependent variables have one-year horizons, I bump the truncation parameter up to 10 for all samples. I note that a truncation parameter of 9 would increase the reported level of significance for γ for PCE growth and CPI inflation from February 2000 to June 2003. Hence, reported statistical significance in Table 5 is conservative.

³¹The differences between the effects of forward guidance across the subsamples are statistically significant at the 1 percent level for PCE growth and the unemployment rate change. This difference is statistically significant at 32 percent level for CPI inflation. See the online Appendix.

TABLE 5—RESPONSES OF MACROECONOMIC VARIABLES TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

Feb. 2000 to May 2006			Feb. 2000 to Jun. 2003			Aug. 2003 to May 2006		
Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2	Funds rate	Forward guidance	R^2
<i>PCE growth results</i>								
0.42 (1.53)	1.56 (3.40)	0.01	0.37 (1.51)	2.88 (2.26)	0.03	1.07 (7.33)	−10.51 (3.82)	0.15
<i>CPI inflation results</i>								
−0.46 (2.04)	8.16 (1.77)	0.13	−0.42 (2.01)	7.06 (3.00)	0.19	−12.94 (21.38)	0.51 (3.44)	0.03
<i>Unemployment change results</i>								
0.70 (1.80)	−5.98 (2.12)	0.09	0.87 (2.01)	−4.01 (1.13)	0.07	−0.05 (1.74)	3.49 (1.24)	0.23
<i>IP growth results</i>								
−4.89 (6.02)	1.65 (9.23)	0.01	−5.65 (6.84)	−9.87 (8.53)	0.04	17.08 (24.92)	−10.10 (4.37)	0.08

Notes: The *Funds rate* columns display the estimates of β_1 , and the *Forward guidance* columns display the estimates of γ from equation (8). Standard errors are shown in parentheses. See the text and the online Appendix for details about estimation and inference.

the online Appendix for $h = 1, \dots, 24$. In general, they show the same pattern of results as in Table 5. In the February 2000 to June 2003 sample, PCE growth and CPI inflation generally respond positively to a forward guidance surprise, while the unemployment rate generally responds negatively. IP growth responds positively at short horizons and is very imprecisely estimated at longer horizons. In the August 2003 to May 2006 sample, PCE growth and IP growth generally respond negatively to a forward guidance surprise, while the unemployment rate generally responds positively. The response of CPI inflation is noisy and imprecisely estimated.

Taken together, the signs of the regression coefficients in Tables 1 through 5 show that forward guidance had different effects from February 2000 to June 2003 than from August 2003 to May 2006. In the early sample when the FOMC was using only economic-outlook language, forward guidance had strong information effects. Forward guidance that caused a surprise increase in the expected funds rate path also caused upward revisions in the outlook of financial market participants and professional forecasters, and this forward guidance was followed by improving economic conditions. However, in the later sample when the FOMC added policy-inclination language, the standard theoretical effects of forward guidance dominated. Forward guidance that caused a surprise increase in the expected funds rate path also caused downward revisions in the outlook of financial market participants and professional forecasters, and this forward guidance was followed by worsening economic conditions.

V. Robustness of Interpretation and Policy Implications

I interpret the previous section as showing that economic-outlook forward guidance from February 2000 to June 2003 had strong information effects but that the addition of policy-inclination language in August 2003 changed the effects of forward guidance to be consistent with standard theory. This interpretation begs

two follow-up questions. First, were there changes in the economy in 2003 other than forward guidance language that could cause similar changes in private sector responses? Second, given that the February 2000 to May 2006 sample has essentially one change to forward guidance, the addition of policy-inclination language from August 2003 to May 2006, what policy implications are we comfortable drawing?

This section addresses these two questions. Subsection VA studies one potential change to the economy in 2003 other than forward guidance language: the drop in the federal funds rate that motivated the FOMC to begin using policy-inclination forward guidance. Subsection VB studies a second potential change to the economy: a change in economic uncertainty from the February 2000 to June 2003 sample to the August 2003 to May 2006 sample. Subsection VC then discusses a change to FOMC forward guidance language on August 9, 2011, and relates this language change to the results in this paper and some policy implications.

A. Changes in the Level of Interest Rates

This subsection studies one potential economic change in 2003 other than forward guidance language that may have changed the private sector's responses to forward guidance: the low level of the federal funds rate. As indicated by Ben S. Bernanke's quotation above, the addition of policy-inclination forward guidance language in August 2003 was not random; it was a response to the low level of interest rates. In particular, some FOMC participants were concerned about losing their traditional policy tool as a result of the ZLB, and the private sector may have shared this concern. Hence, it is not immediately clear if the change in the private sector's responses is coming from the addition of policy-inclination forward guidance or if it is coming from the low level of interest rates and anticipation of hitting the ZLB.

To address this problem, I consider an alternative late sample of June 2004 to May 2006 instead of August 2003 to May 2006. June 2004 was the first meeting during which the FOMC agreed to increase the target federal funds rate. It followed the May 2004 meeting during which Alan Greenspan said, "we have completed our period of accommodation,"³² and after which the FOMC stated, "policy accommodation can be removed." It also followed a June 2004 speech by Kohn (2004) that notes that "[c]ircumstances have changed" from when the FOMC put policy-inclination forward guidance into use. However, as noted above, the FOMC continued to use policy-inclination forward guidance over this sample. Hence, June 2004 to May 2006 represents a sample in which the FOMC was using its traditional policy tool and in which concerns about needing to cut rates to the ZLB had essentially vanished, but policy-inclination forward guidance was still in place.

Table 6 shows select results for the June 2004 to May 2006 sample. The pattern of results in this table is very similar to the results in Tables 1 through 5; however, statistical significance has declined slightly with the smaller sample size.³³ These results indicate that the private sector's responses to policy-inclination forward guidance do not rely on the August 2003 to May 2004 sample when interest rates were lowest and ZLB concerns were present. Hence, it is reasonable to conclude

³² See <https://www.federalreserve.gov/monetarypolicy/files/FOMC20040504meeting.pdf>, page 82.

³³ I use a truncation parameter of 8 to estimate standard errors for the macroeconomic variables.

TABLE 6—RESPONSES OF MACROECONOMIC VARIABLES TO FUNDS RATE AND FORWARD GUIDANCE CHANGES

	Jun. 2004 to May 2006		
	Funds rate	Forward guidance	R^2
<i>Panel A. Stock market</i>			
S&P 500	−9.32 (15.56)	−13.50 (6.20)	0.22
VIX	11.96 (14.44)	10.70 (5.62)	0.18
<i>Panel B. Treasury term premia</i>			
2-year	0.46 (0.40)	0.56 (0.17)	0.41
5-year	−0.48 (0.40)	0.43 (0.23)	0.21
7-year	−0.85 (0.44)	0.41 (0.27)	0.22
10-year	−1.14 (0.52)	0.43 (0.31)	0.23
<i>Panel C. Private-sector borrowing spreads</i>			
BBB − A (3-year)	−1.27 (0.54)	0.39 (0.18)	0.53
BBB − A (10-year)	−0.46 (0.37)	0.26 (0.16)	0.20
OAS	0.64 (0.33)	0.52 (0.17)	0.49
<i>Panel D. Blue chip forecasts</i>			
GDP growth	3.30 (2.93)	−1.32 (0.81)	0.32
CPI inflation	−2.15 (1.08)	−0.01 (0.65)	0.14
Unemployment rate	−0.49 (0.64)	0.60 (0.27)	0.32
<i>Panel E. Macroeconomic variables</i>			
PCE growth	−0.01 (10.13)	−12.88 (5.00)	0.20
CPI inflation	−13.52 (24.93)	−0.19 (5.38)	0.03
Unemployment change	0.38 (2.22)	2.89 (1.39)	0.18
IP growth	24.15 (26.32)	−12.15 (7.40)	0.16

Notes: The *Funds rate* columns display the estimates of β_1 , and the *Forward guidance* columns display the estimates of γ from equation (7). For macroeconomic variables, these estimates are from (8). Standard errors are shown in parentheses. See the text and the online Appendix for details about estimation and inference.

that the policy-inclination language—not the low interest rates and concerns about the ZLB—caused the private sector’s responses to change.

B. *Changes in Uncertainty*

Next, I consider a possible change in economic uncertainty in 2003. The February 2000 to June 2003 sample includes a recession and a generally falling federal funds rate, but the August 2003 to May 2006 sample has no recession and a generally rising federal funds rate. If the early sample has more economic uncertainty than the later sample, then the private sector might anchor its beliefs more to the FOMC's economic outlook in the early sample. This mechanism is similar to Melosi's (2017) and could increase the FOMC's information effects in the early sample independent of the FOMC's language change in August 2003.

Motivated by Melosi (2017), wherein the equilibrium Phillips curve depends on high-order expectations about the output gap and inflation, I study measures of uncertainty related to real activity and inflation. The first two measures are of real and macroeconomic uncertainty from Ludvigson, Ma, and Ng (2019) and Jurado, Ludvigson, and Ng (2015). Real uncertainty is measured as a weighted average of the root expected forecast error for 73 real activity variables. Macroeconomic uncertainty is similar but also includes data on prices and financial variables. The third and fourth measures are of GDP growth and inflation uncertainty from Rossi and Sekhposyan (2015). These measures are based on the size of forecast errors from the Survey of Professional Forecasters. I test the differences in these four uncertainty levels across my two samples in the online Appendix and find no clear evidence that overall economic uncertainty changed from the early sample to the late sample. Real uncertainty was essentially constant across both samples. While macroeconomic uncertainty was lower in the late sample, this change is not statistically significant. GDP growth uncertainty fell from my early sample to my late sample, but inflation uncertainty rose from the early sample to the late sample.³⁴ This lack of clear evidence that uncertainty either rose or fell between my sample periods is consistent with Jurado, Ludvigson, and Ng's (2015) findings that 1974–1975, 1980, and 2008–2009 were the only big episodes of uncertainty. These results suggest that fluctuations in uncertainty are unlikely to have made a material change in the FOMC's information effects.

C. *The FOMC's Language Change in 2011*

The previous two subsections support my interpretation that the addition of policy-inclination forward guidance in August 2003 materially changed the private sector's responses to forward guidance. However, caution is warranted when drawing policy implications from this one policy change. To bolster the results in this paper, I briefly discuss some of the literature that has studied FOMC communication since the start of the ZLB period in December 2008. I especially focus on the FOMC's language change in its August 9, 2011, statement and highlight the similarities with August 2003 language change.

Upon hitting the ZLB in December 2008, the FOMC stated, “weak economic conditions are likely to warrant exceptionally low levels of the federal funds rate

³⁴This increase in inflation uncertainty from 2000 to 2006 is consistent with Figure 2 of Söderlind (2011), who measures uncertainty from professional forecasters' subjective uncertainty distributions.

for some time.” It changed this guidance to “an extended period” in March 2009. However, as I note above, adding or changing policy-inclination language may not represent a policy change if the new language simply tells the private sector what it already knows, and it is not clear that the December 2008 or March 2009 language changes represent policy changes.³⁵ Rather, the literature highlights the August 2011 FOMC meeting as having an important change in policy-inclination language.

To establish that the August 2003 language change was a material change in policy, I showed that two conditions were met. First, in Section I, I showed that the FOMC intended to use policy-inclination forward guidance as an additional means of stimulus. Second, in Section IIB, I showed that the federal funds futures market became less volatile, consistent with the FOMC communicating its actual policy inclinations. The FOMC’s language change in August 2011 also meets these two conditions.

At its August 2011 meeting, the FOMC began using calendar-based policy-inclination language, stating “that economic conditions ... are likely to warrant exceptionally low levels for the federal funds rate at least through mid-2013.” The minutes for the August 2011 meeting note that some FOMC participants viewed this “stronger forward guidance as a step in the direction of additional accommodation.”³⁶ In a subsequent speech, Bernanke (2011) indicated that this refining of forward guidance was a monetary stimulus tool. Hence, this change in language meets the condition that the FOMC intended it to be accommodative.

The financial markets believed that the FOMC was communicating its actual policy inclinations and the FOMC appeared to follow through on its forward guidance. Measures of interest rate volatility, including the 2-year-ahead swaption-implied volatility and 2-year-ahead overnight interest-rate swaps, fell on August 9, 2011 (Woodford 2012; Femia, Friedman, and Sack 2013). Following this meeting, Swanson and Williams (2014) note that 1- and 2-year Treasury yields became unresponsive to macroeconomic news, and they tie this result to the August 2011 language change. Andrade et al. (2019) show that disagreement about 1- and 2-year-ahead interest rate forecasts fell to historic lows.

Other financial market responses to the August 2011 language change are consistent with standard theory. 4-, 8-, and 12-quarter-ahead Eurodollar futures immediately dropped, while the S&P 500 jumped 4.7 percent (Femia, Friedman, and Sack 2013). I note that no changes to the current federal funds rate or to LSAP policy were announced, no press conference was held, and no SEPs were provided. The immediate financial market responses were to the FOMC statement language only. Consistent with this, Swanson (2017) finds a large forward guidance response from futures markets, but essentially no LSAP response. With regard to the macroeconomic effects of this language change, Campbell et al. (2017) find that FOMC communication “achieved more favorable outcomes” beginning in late 2011.

These results for the August 2011 language change and my results for the August 2003 language change have consistent policy implications: when the FOMC

³⁵ At the March 2009 FOMC meeting, James Bullard noted “the fact that [the fed funds rate is] going to stay at zero for a long time is not news right now.” See <https://www.federalreserve.gov/monetarypolicy/files/FOMC20090318meeting.pdf>. Swanson and Williams (2014) and Campbell et al. (2017) argue that financial markets viewed liftoff from the ZLB as only a few quarters away in 2009 and 2010.

³⁶ See <https://www.federalreserve.gov/monetarypolicy/fomcminutes20110809.htm>.

intended to change its policy stimulus via policy-inclination forward guidance and when that forward guidance conveyed the FOMC's actual policy inclinations, then the FOMC has been perceived as credible and standard theoretical results have applied. Both the August 2003 and August 2011 language changes also suggest that policy-inclination forward guidance can help manage the economy when interest rates are low and pegged.

VI. Conclusions

From February 2000 to June 2003, FOMC statements included only forward guidance about the economic outlook. In contrast, the statements from August 2003 to May 2006 also included forward guidance about the FOMC's policy inclinations. I provide evidence that the FOMC intended for this policy-inclination language to have effects consistent with standard macroeconomic theory, and that the FOMC was conveying its actual policy inclinations. I then use the August 2003 language change to test if the nature of forward guidance language can affect how the private sector responds to forward guidance. From February 2000 to June 2003, I find that forward guidance that reduced the expected path of the federal funds rate caused an increase in term premia, a decrease in stock prices, a decrease in expected GDP growth, and a subsequent increase in the unemployment rate. From August 2003 to May 2006, forward guidance had the opposite effects.

These results show that financial markets, private forecasters, and the macroeconomy can react differently to different aspects of forward guidance. When the FOMC uses only economic-outlook forward guidance, the effects of forward guidance are consistent with strong information effects: a negative economic outlook causes the private sector to revise down its expectations of the economy as interest rates fall. Economic conditions subsequently worsen. When the FOMC also uses policy-inclination forward guidance, standard macroeconomic theory may apply: forward guidance that decreases the funds rate path causes the private sector to revise up its expectations of the economy. Economic conditions subsequently improve. Overall, the effects of forward guidance can depend on the FOMC's choice to use one or both of the economic-outlook and policy-inclination aspects of forward guidance.

While my sample period does not have a zero lower bound episode, it does have a period of low and mostly constant rates. Hence, these results shed light on how the private sector reacts to changes in forward guidance language with pegged interest rates. Further, the results for the August 2003 to May 2006 sample show that the standard effects of forward guidance can apply to an interest rate peg and the subsequent lift-off, suggesting that forward guidance can be a useful tool for managing ZLB episodes.

Finally, the results of this paper are applicable to current policy discussions. The FOMC has continued to use forward-looking language about both the economic outlook and its policy inclinations. Further, the FOMC has begun releasing with its statements summaries of economic projections, which include FOMC participants' forecasts of macroeconomic variables as well as their views on the appropriate path of the federal funds rate. While these projections provide a different means of communication than FOMC statement language, the theory underpinning their effects

remains the same. Changes in FOMC forecasts of macroeconomic variables may operate through an information channel, causing interest rates, stock prices, and expected GDP growth to move in the same direction. In contrast, changes in projections of the appropriate federal funds rate path, the much watched “dot plots,” may have effects more consistent with standard macroeconomic theory, causing interest rates to move in the opposite direction of stock prices without puzzling revisions to inflation and GDP forecasts.

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